SSC MTS 2024 question paper with answers

16 October 2024 · Shift 3

80 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 10 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 16 questions

Q.1

Lokesh received an amount of ₹10,500 in 5 years, including simple interest, on a principal amount of ₹7,500. What would be the total amount he receives if the interest rate is increased by 4%?

(A) ₹12,595
(B) ₹12,550
(C) ₹12,000
(D) ₹11,500

Q.2

If the height of a triangle is increased by 20% and the base is increased by 12%. What is the new area (in cm^2) if the old area of the triangle was 200 cm^2?

(A) 234.4
(B) 216.7
(C) 268.8
(D) 254.4

Q.4

An item is listed for ₹450 with 15% trade discount. Find the amount of trade discount.

(A) ₹382.50
(B) ₹517.50
(C) ₹67.50
(D) ₹52.50

Q.5

The management batch of a college has two sections, A and B. Section A has 22 students and their average marks are 75, whereas section B has 30 students and their average marks are 60. What are the average marks of the whole batch? (Round off to two decimal places)

(A) 65.23
(B) 58
(C) 66.35
(D) 59.34

Q.6

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 4% per year rate of interest for 6 years.

(A) 680
(B) 760
(C) 720
(D) 660

Q.7

A solid cylinder has a total surface area of 352 cm^2. Its curved surface area is one-eighth of the total surface area. The radius (in cm) of the cylinder is: [Use $\pi = 22/7$]

(A) 11
(B) 8
(C) 7
(D) 9

Q.8

24 boys and a certain number of girls appeared for an exam. The average score of the boys was 42 and the average score of the girls was 34, while the combined average score was 38. How many girls appeared for the exam?

(A) 28
(B) 30
(C) 24
(D) 26

Q.9

Find the least number which when divided by 20,25,35 and 40 leaves remainders 14,19,29and 34, respectively.

(A) 1394
(B) 1384
(C) 1284
(D) 1294

Q.10

What should be subtracted from 30, 56, 40 and 76 so that the remaining numbers may be proportional?

(A) 10
(B) 4
(C) 8
(D) 6

Q.11

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 4% per year rate of interest for 2 years.

(A) 280
(B) 180
(C) 200
(D) 240

Q.12

The average of 17 results is 62. The average of the first eighth is 48 and that of the last eighth is 84, the ninth result is:

(A) 2.5
(B) –1.5
(C) –2
(D) 2

Q.13

The cost price of 12 bananas is equal to the sale price of 10 bananas. What is the percentage of profit?

(A) 16%
(B) 12%
(C) 24%
(D) 20%

Q.14

In a jeep driving at a speed of 9 m/s, a thief is fleeing on a straight road. He is being pursued by a police officer riding a motorbike at a speed of 10 m/s. How long would it take a police officer to apprehend the burglar if there is a 100 m instantaneous distance between the vehicle and the motorbike?

(A) 95 s
(B) 98 s
(C) 100 s
(D) 102 s

Q.16

In an examination, there are four subjects of 100 marks each. A student scores 78% in the first subject, 75% in the second subject and 90% in the third subject. She scored 85% in aggregate. Her percentage of marks in the fourth subject is:

(A) 97%
(B) 95%
(C) 87%
(D) 81%

Q.18

A train travels at a speed of 120 km/h. How many metres will it travel in 60 minutes?

(A) 140000
(B) 120000
(C) 145000
(D) 135000

Q.20

Rajneesh drove his car at a speed of 99 km/h for 2 hours and then he took a rest for one hour. Then, he further drove for 3 hours at a speed of 60 km/h. What was Rajneesh's average speed (in km/h) during his entire journey?

(A) 69
(B) 65
(C) 67
(D) 63

Reasoning Ability and Problem Solving 16 questions

Q.1

LIFE is related to NKHG in a certain way based on the English alphabetical order. In the same way, SLOPE is related to UNQRG. To which of the following is DUAL related, following the same logic?

(A) FWCN
(B) FXDN
(C) FXCN
(D) FWDN

Q.2

394 is related to 242 following a certain logic. Following the same logic, 461 is related to 309. To which of the following is 428 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 282
(B) 257
(C) 261
(D) 276

Q.4

In a certain code language, 'JARS' is coded as '3175' and 'SIDE' is coded as '4832'. What is the code for 'S' in the given code language?

(A) 3
(B) 4
(C) 8
(D) 5

Q.5

In a certain code language, 'bring cold water' is coded as 'qh pw ut' and 'water is pure' iscoded as 'mk zv qh'. How is 'water' coded in the given language?

(A) pw
(B) qh
(C) ut
(D) zv

Q.6

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). C is sitting second to the left of A. Only one person is sitting between B and C when counted from right of B. E is sitting third to the right of F. D is sitting to the immediate left of E. The position of how many persons excluding A will remain unchanged if all the persons are arranged in the English alphabetical order in clockwise direction, starting from A, keeping the position of A unchanged?

(A) One
(B) Three
(C) None
(D) Two

Q.8

In a certain code language, 'POT' is coded as '15-14-19' and 'SPLIT' is coded as '18-15-11-8-19'. What is the code for 'PERFORM' in the given code language?

(A) 15-4-17-5-14-17-13
(B) 15-4-17-5-14-17-12
(C) 15-4-17-6-14-17-12
(D) 15-4-17-5-15-17-12

Q.9

What should come in place of the question mark (?) in the given series based on the English alphabetical order? BFL, HLR, NRX, TXD, ?

(A) JDZ
(B) ZDJ
(C) DZJ
(D) ZJD

Q.10

What will come in place of the question mark (?) in the following equation if '+' and '–' are interchanged and '×' and '÷' are interchanged? 60 × 12 – 6 ÷ 4 + 15 = ?

(A) 12
(B) 14
(C) 11
(D) 13

Q.11

In a certain code language, 'WILD' is coded as '3197' and 'LOVE' is coded as '2346'. What is the code for 'L' in that language?

(A) 3
(B) 2
(C) 9
(D) 6

Q.12

What will come in the place of the question mark (?) in the following equation if '–' and '+' are interchanged and '÷' and '×' are interchanged? 13 ÷ 4 + 16 × 2 – 21 = ?

(A) 65
(B) 59
(C) 42
(D) 61

Q.13

This question has two statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which of the given conclusions logically follow(s) from the given statements. Statements: Some hoops are cows. All cows are bulls. Conclusions: (I) Some bulls are hoops. (II) Some hoops are bulls.

(A) Only conclusion (II) follows
(B) Only conclusion (I) follows
(C) Both conclusions (I) and (II) follow
(D) Neither conclusion (I) nor (II) follows

Q.14

13 is related to 104 following a certain logic. Following the same logic, 17 is related to 136. To which ofthe following is 23 related, following the same logic?(NOTE: Operations should be performed on the whole numbers, without breaking down the numbersinto its constituent digits. E.g. 13 – Operations on 13 such as adding /subtracting /multiplying etc. to 13can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1and 3 is not allowed)

(A) 184
(B) 92
(C) 207
(D) 138

Q.15

P, Q, R, S, T, U and V are sitting around a circular table facing the centre (but not necessarily in the same order). T sits to the immediate right of V. S sits second to the right of Q. U is an immediate neighbour of both V and S. P sits second to the left of U. Who is sitting to the immediate left of R?

(A) P
(B) S
(C) Q
(D) T

Q.18

HP21 is related to XF34 in a certain way. In the same way, KS72 is related to AI85. To which of the following is NV39 related, following the same logic?

(A) LD52
(B) DL55
(C) LD55
(D) DL52

Q.19

What should come in place of the question mark (?) in the given series? 3 16 66 201 406 ?

(A) 412
(B) 409
(C) 410
(D) 411

Q.20

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statement(s) is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: No chair is a recliner. No recliner is a sofa. No recliner is a mat. Conclusions: (I) No chair is a sofa. (II) Some mats are sofa.

(A) Neither conclusion (I) nor (II) follows.
(B) Only conclusion (II) follows.
(C) Only conclusion (I) follows.
(D) Both conclusions (I) and (II) follow.

General Awareness 24 questions

Q.1

The Shuddhi Movement to reconvert to the Hindu fold the converts to other religions, was started by which of the following organisations?

(A) Arya Samaj
(B) Satyashodhak Samaj
(C) Brahmo Samaj
(D) Dharma Sabha

Q.2

Who is the author of ‘Temple Tales: Secrets & Stories of India’s Sacred Places’?

(A) Sudha G Tilak
(B) Shashi Tharoor
(C) Malathi Ramachandran
(D) Mulk Raj Anand

Q.3

Who among the following is the author of the book named 'Streepurushtulna'?

(A) Pandit Ramabai
(B) Rashasundari Devi
(C) Tarabai Shinde
(D) Sarojini Naidu

Q.4

From which of the following countries has the Indian Constitution borrowed the procedure of making amendments in the Constitution?

(A) South Africa
(B) Australia
(C) Ireland
(D) Russia

Q.5

Which of the following was NOT found in the Harappan excavation sites?

(A) Dwelling houses
(B) Public hospital
(C) Public buildings
(D) Public bath

Q.6

The Constitution of India has borrowed many provisions from different countries as well as from the Government of India Act of _____.

(A) 1942
(B) 1935
(C) 1949
(D) 1919

Q.8

Which of the following Union Territories has the lowest population as per the Census of India 2011?

(A) Lakshdweep
(B) Delhi
(C) Andaman and Nicobar
(D) Daman and Diu

Q.9

Squash player Anahat Singh belongs to which city in India?

(A) Haryana
(B) Mumbai
(C) Bengaluru
(D) Delhi

Q.10

Which of the following is considered unicellular green algae?

(A) Cladophora
(B) Oedogonium
(C) Chlorophyta
(D) Chlamydomonas

Q.11

The structure of methanol is CH_3OH. This kind of representation is known as _____.

(A) bond-line structural representation
(B) wedge-and-dash representation
(C) complete structural formula
(D) condensed structural formula

Q.12

Which of the following pollutants is responsible for the acidification of ocean?

(A) Carbon monoxide (CO)
(B) Carbon dioxide (CO_2)
(C) Nitrogen Dioxide (NO_2)
(D) Sulphur dioxide (SO_2)

Q.13

The Integrated Goods and Services Tax (Amendment) Bill, 2023 was introduced in Lok Sabha on 11 August 2023, which amends the _____.

(A) Integrated Goods and Services Tax (IGST) Act, 2015
(B) Integrated Goods and Services Tax (IGST) Act, 2016
(C) Integrated Goods and Services Tax (IGST) Act, 2017
(D) Integrated Goods and Services Tax (IGST) Act, 2014

Q.14

Which raga was composed by music emperor Mian Tansen to bring rain?

(A) Megh Malhar
(B) Megh Pushpa
(C) Mayurapriya
(D) Sharvari Tarang

Q.15

Muscle cells and muscle fibres are developed from embryonic precursor cells known as:

(A) myonuclei
(B) myocyte
(C) myoblasts
(D) myofibril

Q.16
VP Dhananjayan and Shanta Dhananjayan were honoured with Rukminidevi Memorial Award in 2009 for which of the following Indian classical dance forms?

(A) Bharatanatyam
(B) Odissi
(C) Mohiniyattam
(D) Sattriya

Q.17
Which of the following temples is NOT an example of Nagara style of architecture?

(A) Sun temple, Konark
(B) Lingaraja temple, Bhubaneshwar
(C) Kandariya temple, Madhya Pradesh
(D) Kailasha temple, Kanchipuram

Q.18
Which of the following is the headquarter of East Central Railway?

(A) Hajipur
(B) Guwahati
(C) Kanpur
(D) Kolkata

Q.19
Lachhu Maharaj has been associated with which dance form?

(A) Odissi
(B) Manipuri
(C) Bharatanatyam
(D) Kathak

Q.20
In September 2023, at the BRICS Innovation Forum in Moscow, who among the following, who is a Chief Innovation Officer from Telangana, received the World Innovation Award?

(A) T Harish Kumar
(B) Shanta Thoutam
(C) KT Rama Rao
(D) Kalvakuntla Chandrashekar Rao

Q.21
Which of the following is NOT a religious festival?

(A) Good Friday
(B) Gandhi Jayanti
(C) Guru Purab
(D) Muharram

Q.22
According to the Census of India 2001, how much percentage of the immigrant population was from Sri Lanka?

(A) 8.1%
(B) 1.6%
(C) 7.1%
(D) 2.9%

Q.23
To which state does Indian shooter Abhinav Bindra belong?

(A) Rajasthan
(B) Madhya Pradesh
(C) Punjab
(D) Uttarakhand

Q.24
According to Koeppen's Scheme of climatic regions of India, in which of the following states is the 'BWhw' and 'BShw' type climate found?

(A) Maharashtra
(B) Rajasthan
(C) Bihar
(D) Andhra Pradesh

Q.25
Dhamma is the _____ word for the Sanskrit word 'Dharma'.

(A) Tamil
(B) Pali
(C) Abahatta
(D) Prakrit

English Language and Comprehension 24 questions

Q.1

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Rahul speaks fluently English / although he makes / quite a lot of mistakes /while writing.

(A) while writing
(B) quite a lot of mistakes
(C) although he makes
(D) Rahul speaks fluently English

Q.2

Select the option that can be used as a one-word substitute for the given group of words. A person or entity that is unable to pay debts.

(A) Recluse
(B) Busted
(C) Insurgent
(D) Insolvent

Q.3

Select the correctly spelt word to fill in the blank and make the sentence meaningful. Sheetal has been studying engineering for five years, _____ countless hours to her coursework.

(A) deddicating
(B) daidicating
(C) dedecating
(D) dedicating

Q.4

Select the option that rectifies the underlined spelling error. The impacts of climate change on different sectors of society are “interelated”.

(A) interrilated
(B) enterrelated
(C) interrelated
(D) interreleted

Q.5

Select the most appropriate option to fill in the blanks. His dog did not _____ when thieves were slashing the _____ of his precious tree.

(A) claw
(B) stem
(C) shout
(D) bark

Q.6

Select the sentence that has a grammatical error.

(A) I wish I had eraser to erase all the mistakes in my life.
(B) A young boy offered her his seat.
(C) I have a lot of work to complete.
(D) This is a very difficult task to accomplish.

Q.7

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. We / enjoyed lunch / he gave / us.

(A) We
(B) enjoyed lunch
(C) us
(D) he gave

Q.8

Select the most appropriate option to fill in the blank. If you don’t have a lump sum to _____ in a fixed deposit, you can surely opt for a long-term recurring deposit.

(A) gather
(B) publish
(C) draw
(D) invest

Q.9

Select the most appropriate ANTONYM of the underlined word. The meteorological department predicted “heavy” rain over the coming days.

(A) Plentiful
(B) Drizzly
(C) Rare
(D) Light

Q.10

Select the most appropriate ANTONYM of the underlined word in the given sentence. As the orchestra played, the audience was “mesmerised” by the beautiful melodies.

(A) turbid
(B) alienated
(C) dirty
(D) abstracted

Q.11

Select the appropriate option that can substitute the underlined word in the following sentence to make it grammatically correct. Only “the beautiful” painting is to be displayed in the exhibition.

(A) beautiful
(B) the much beautiful
(C) more beautiful
(D) the most beautiful

Q.12

Parts of the following sentence have been given as options. Select the option that contains an error. Please bring two bananas and the piece of cake from the market.

(A) Please bring
(B) from the market
(C) two bananas
(D) and the piece of cake

Q.13

Select the most appropriate option to fill in the blank. He couldn't attend the meeting _____ he was sick.

(A) although
(B) so
(C) but
(D) because

Q.14

Select the most appropriate meaning of the given idiom. Green thumb

(A) To visit the vegetable market
(B) Being good at growing plants
(C) Indulging in corruption
(D) Preparing for war

Q.15

Select the most appropriate synonym of the given word. Fatigued

(A) Furious
(B) Tired
(C) Famous
(D) Tricky

Q.17

Select the most appropriate synonym of the given word. Proficient

(A) Recalcitrant
(B) Accomplished
(C) Glamorous
(D) Callous

Q.18

Select the INCORRECTLY spelt word in the following sentence. His performace in the play was outstanding.

(A) play
(B) outstanding
(C) performace
(D) His

Q.19

Select the most appropriate meaning of the underlined idiom. Despite their differences, the counties found common ground and “buried the hatchet”.

(A) To continue a conflict
(B) To seek revenge in the future
(C) To settle a dispute peacefully
(D) To disagree publicly

Q.20

Select the option that can be used as a one-word substitute for the given group of words. Exactly the same words as were used originally.

(A) Verbatim
(B) Ornate
(C) Precise
(D) Factual

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Sniffer dog Tucker uses (1)_____ nose to help researchers find out why a killer whale population off the northwest coast of the United States is (2)_____ the decline. He searches for whale faeces floating on the surface of the water, which are then collected for examination. He is one of the elite team members of detection dogs used by scientists studying a (3)_____ of species including right whales and killer whales. Conservation canines are fast becoming indispensable tools for biologists according to Aimee Hurt, associate director and co-founder of Working Dogs for Conservation, (4)_____ in Three Forks, Montana. Over the last few years, though, so many new conservation dog projects have sprung up that Hurt can no (5)_____ keep track of them all. Her organisation’s dogs and their handlers are fully booked to assist field researchers into 2012.

Choose the most appropriate option to fill in blank no. 1.

(A) him
(B) its
(C) his
(D) her

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Sniffer dog Tucker uses (1)_____ nose to help researchers find out why a killer whale population off the northwest coast of the United States is (2)_____ the decline. He searches for whale faeces floating on the surface of the water, which are then collected for examination. He is one of the elite team members of detection dogs used by scientists studying a (3)_____ of species including right whales and killer whales. Conservation canines are fast becoming indispensable tools for biologists according to Aimee Hurt, associate director and co-founder of Working Dogs for Conservation, (4)_____ in Three Forks, Montana. Over the last few years, though, so many new conservation dog projects have sprung up that Hurt can no (5)_____ keep track of them all. Her organisation’s dogs and their handlers are fully booked to assist field researchers into 2012.

Choose the most appropriate option to fill in blank no. 2.

(A) on
(B) off
(C) after
(D) at

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Sniffer dog Tucker uses (1)_____ nose to help researchers find out why a killer whale population off the northwest coast of the United States is (2)_____ the decline. He searches for whale faeces floating on the surface of the water, which are then collected for examination. He is one of the elite team members of detection dogs used by scientists studying a (3)_____ of species including right whales and killer whales. Conservation canines are fast becoming indispensable tools for biologists according to Aimee Hurt, associate director and co-founder of Working Dogs for Conservation, (4)_____ in Three Forks, Montana. Over the last few years, though, so many new conservation dog projects have sprung up that Hurt can no (5)_____ keep track of them all. Her organisation's dogs and their handlers are fully booked to assist field researchers into 2012.

Choose the most appropriate option to fill in blank no. 3.

(A) number
(B) division
(C) cadre
(D) cacophony

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Sniffer dog Tucker uses (1)_____ nose to help researchers find out why a killer whale population off the northwest coast of the United States is (2)_____ the decline. He searches for whale faeces floating on the surface of the water, which are then collected for examination. He is one of the elite team members of detection dogs used by scientists studying a (3)_____ of species including right whales and killer whales. Conservation canines are fast becoming indispensable tools for biologists according to Aimee Hurt, associate director and co-founder of Working Dogs for Conservation, (4)_____ in Three Forks, Montana. Over the last few years, though, so many new conservation dog projects have sprung up that Hurt can no (5)_____ keep track of them all. Her organisation's dogs and their handlers are fully booked to assist field researchers into 2012.

Choose the most appropriate option to fill in blank no. 4.

(A) rested
(B) living
(C) surviving
(D) based

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Sniffer dog Tucker uses (1)_____ nose to help researchers find out why a killer whale population off the northwest coast of the United States is (2)_____ the decline. He searches for whale faeces floating on the surface of the water, which are then collected for examination. He is one of the elite team members of detection dogs used by scientists studying a (3)_____ of species including right whales and killer whales. Conservation canines are fast becoming indispensable tools for biologists according to Aimee Hurt, associate director and co-founder of Working Dogs for Conservation, (4)_____ in Three Forks, Montana. Over the last few years, though, so many new conservation dog projects have sprung up that Hurt can no (5)_____ keep track of them all. Her organisation's dogs and their handlers are fully booked to assist field researchers into 2012.

Choose the most appropriate option to fill in blank no. 5.

(A) shorter
(B) lesser
(C) moreover
(D) longer

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.2 **(C)**	Q.4 **(C)**	Q.5 **(C)**	Q.6 **(C)**	Q.7 **(C)**
Q.8 **(C)**	Q.9 **(A)**	Q.10 **(B)**	Q.11 **(D)**	Q.12 **(C)**	Q.13 **(D)**
Q.14 **(C)**	Q.16 **(A)**	Q.18 **(B)**	Q.20 **(D)**		

Reasoning Ability and Problem Solving

Q.1 **(A)**	Q.2 **(D)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(A)**	Q.8 **(B)**
Q.9 **(B)**	Q.10 **(B)**	Q.11 **(A)**	Q.12 **(A)**	Q.13 **(C)**	Q.14 **(A)**
Q.15 **(D)**	Q.18 **(D)**	Q.19 **(D)**	Q.20 **(A)**		

General Awareness

Q.1 **(A)**	Q.2 **(A)**	Q.3 **(C)**	Q.4 **(A)**	Q.5 **(B)**	Q.6 **(B)**
Q.8 **(A)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(B)**	Q.13 **(C)**
Q.14 **(A)**	Q.15 **(C)**	Q.16 **(A)**	Q.17 **(D)**	Q.18 **(A)**	Q.19 **(D)**
Q.20 **(B)**	Q.21 **(B)**	Q.22 **(D)**	Q.23 **(D)**	Q.24 **(B)**	Q.25 **(D)**

English Language and Comprehension

Q.1 **(D)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(A)**
Q.7 **(B)**	Q.8 **(D)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(D)**	Q.12 **(D)**
Q.13 **(D)**	Q.14 **(B)**	Q.15 **(B)**	Q.17 **(B)**	Q.18 **(C)**	Q.19 **(C)**
Q.20 **(A)**	Q.21 **(C)**	Q.22 **(A)**	Q.23 **(A)**	Q.24 **(D)**	Q.25 **(D)**

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) ₹12,000

SI = 10,500 − 7,500 = ₹3,000, so the rate is 3,000 × 100/(7,500 × 5) = 8%. At 8% + 4% = 12%, SI = 7,500 × 12 × 5/100 = ₹4,500 and the amount is ₹12,000.

Q.2 · Answer (C) 268.8

Area is half the base times the height, so it is multiplied by 1.2 × 1.12 = 1.344: 200 × 1.344 = 268.8 cm².

Q.4 · Answer (C) ₹67.50

Trade discount = 15% of ₹450 = ₹67.50.

Q.5 · Answer (C) 66.35

(22 × 75 + 30 × 60)/52 = (1650 + 1800)/52 = 3450/52 ≈ 66.35.

Q.6 · Answer (C) 720

SI = 3000 × 4 × 6/100 = ₹720.

Q.7 · Answer (C) 7

The curved surface is 352/8 = 44 cm², so the two ends take 352 − 44 = 308 cm²: $2\pi r^2 = 308$, $\pi r^2 = 154$ and $r^2 = 49$ ($\pi = 22/7$), so r = 7 cm.

Q.8 · Answer (C) 24

With g girls: 24 × 42 + 34g = 38(24 + g), so 1008 + 34g = 912 + 38g, 4g = 96 and g = 24.

Q.9 · Answer (A) 1394

Each remainder is 6 less than its divisor (20 − 14 = 25 − 19 = 35 − 29 = 40 − 34 = 6), so the number is LCM(20, 25, 35, 40) − 6 = 1400 − 6 = 1394.

Q.10 · Answer (B) 4

Subtracting 4 leaves 26, 52, 36 and 72, and 26 : 52 = 36 : 72 = 1 : 2.

Q.11 · Answer (D) 240

SI = 3000 × 4 × 2/100 = ₹240.

Q.12 · Answer (C) −2

The total is 17 × 62 = 1054. The first eight total 8 × 48 = 384 and the last eight 8 × 84 = 672, so the ninth is 1054 − 384 − 672 = −2.

Q.13 · Answer (D) 20%

Selling 10 bananas brings in the cost of 12: a profit of 2 on a cost of 10, or 20%.

Q.14 · Answer (C) 100 s

The motorbike gains 10 − 9 = 1 m/s on the jeep, so it closes the 100 m in 100 seconds.

Q.16 · Answer (A) 97%

85% of 400 is 340 marks in all. 78 + 75 + 90 = 243, so the fourth subject gives 340 − 243 = 97 marks out of 100: 97%.

Q.18 · Answer (B) 120000

60 minutes is 1 hour, so the train covers 120 km = 120000 m.

Q.20 · Answer (D) 63

Distance = 99 × 2 + 60 × 3 = 198 + 180 = 378 km. The time, rest included, is 2 + 1 + 3 = 6 hours: 378/6 = 63 km/h.

Reasoning Ability and Problem Solving

Q.1 · Answer (A) FWCN

Each letter moves 2 places forward: L → N, I → K, F → H, E → G. So D → F, U → W, A → C, L → N: FWCN.

Q.2 · Answer (D) 276
The second number is 152 less than the first: 394 − 152 = 242 and 461 − 152 = 309. Likewise 428 − 152 = 276.

Q.4 · Answer (A) 3
S is the only letter in both words and 3 the only digit in both codes, so S = 3.

Q.5 · Answer (B) qh
'water' is the only word in both sentences and 'qh' the only code in both, so water = qh.

Q.6 · Answer (A) One
Facing the centre, each person's left is clockwise. Clockwise from A the seats are A, F, C, G, B, E, D, which fits every clue (C second to A's left, one person between B and C counting from B's right, E third to F's right, D to E's immediate left). In alphabetical order clockwise from A they would hold A, B, C, D, E, F, G, so only C keeps its place.

Q.8 · Answer (B) 15-4-17-5-14-17-12
Each letter is coded as its place in the alphabet minus 1: P (16) → 15, O (15) → 14, T (20) → 19. So PERFORM → 15-4-17-5-14-17-12.

Q.9 · Answer (B) ZDJ
Each letter moves 6 places forward from one cluster to the next, wrapping round after Z: T → Z, X → D, D → J: ZDJ.

Q.10 · Answer (B) 14
After interchanging the signs: 60 ÷ 12 + 6 × 4 − 15 = 5 + 24 − 15 = 14.

Q.11 · Answer (A) 3
L is the only letter in both words and 3 the only digit in both codes, so L = 3.

Q.12 · Answer (A) 65
After interchanging the signs: 13 × 4 − 16 ÷ 2 + 21 = 52 − 8 + 21 = 65.

Q.13 · Answer (C) Both conclusions (I) and (II) follow
The hoops that are cows are also bulls, so some hoops are bulls (II), and so some bulls are hoops (I). Both follow.

Q.14 · Answer (A) 184
The second number is 8 times the first: 13 × 8 = 104 and 17 × 8 = 136. Likewise 23 × 8 = 184.

Q.15 · Answer (D) T
Going round the table to each person's left from U: U, S, P, Q, R, T, V. Then T is to V's immediate right, S second to Q's right, U between V and S and P second to U's left; T sits to R's immediate left.

Q.18 · Answer (D) DL52
Each letter moves 16 places forward (wrapping round after Z) and the number rises by 13: H → X, P → F, 21 → 34; K → A, S → I, 72 → 85. So N → D, V → L and 39 → 52: DL52.

Q.19 · Answer (D) 411
Multiply by 5, 4, 3, 2, 1 and add 1, 2, 3, 4, 5: 3 × 5 + 1 = 16, 16 × 4 + 2 = 66, 66 × 3 + 3 = 201, 201 × 2 + 4 = 406, 406 × 1 + 5 = 411.

Q.20 · Answer (A) Neither conclusion (I) nor (II) follows.
The statements only keep chairs, sofas and mats apart from recliners; they say nothing about how chairs, sofas and mats relate to one another. So neither conclusion follows.

General Awareness

Q.1 · Answer (A) Arya Samaj
The Arya Samaj, founded by Swami Dayananda Saraswati in 1875, ran the Shuddhi movement to bring converts back into the Hindu fold.

Q.2 · Answer (A) Sudha G Tilak
'Temple Tales: Secrets and Stories from India's Sacred Places' (2019), a book for children, is by the journalist Sudha G. Tilak.

Q.3 · Answer (C) Tarabai Shinde
Tarabai Shinde wrote 'Stree Purush Tulana' (1882), a sharp critique of how society treated women.

Q.4 · Answer (A) South Africa
The procedure for amending the Constitution, like the election of members of the Rajya Sabha, was borrowed from the Constitution of South Africa.

Q.5 · Answer (B) Public hospital
Harappan sites have yielded dwelling houses, public buildings such as granaries, and the Great Bath at Mohenjo-daro, but no public hospital.

Q.6 · Answer (B) 1935
Much of the Constitution, including the federal scheme, the office of Governor, the Public Service Commissions and the emergency provisions, came from the Government of India Act, 1935.

Q.8 · Answer (A) Lakshdweep
Lakshadweep, with 64,473 people in 2011, is the least populous Union Territory.

Q.9 · Answer (D) Delhi
Anahat Singh, born in 2008, is a squash player from Delhi; she was the youngest member of India's team at the 2022 Commonwealth Games.

Q.10 · Answer (D) Chlamydomonas
Chlamydomonas is a single-celled green alga that swims with two flagella. Cladophora and Oedogonium are many-celled filaments, and Chlorophyta is the name of the whole green-algae group.

Q.11 · Answer (D) condensed structural formula
A condensed structural formula writes each carbon with its attached atoms in a row, as in CH_3OH, instead of drawing every bond.

Q.12 · Answer (B) Carbon dioxide (CO_2)
Carbon dioxide absorbed by sea water forms carbonic acid, which lowers the ocean's pH.

Q.13 · Answer (C) Integrated Goods and Services Tax (IGST) Act, 2017
The Integrated Goods and Services Tax Act dates from 2017, when GST began; the 2023 Bill amended it to tax online gaming supplied from outside India.

Q.14 · Answer (A) Megh Malhar
SSC's key gives Megh Malhar. Ragas of the Malhar family are sung in the monsoon and said to bring rain, and tradition credits Tansen with one of them, Miyan ki Malhar; of the options only Megh Malhar is a Malhar raga.

Q.15 · Answer (C) myoblasts
Myoblasts are embryonic precursor cells that fuse to form the long, many-nucleated fibres of skeletal muscle.

Q.16 · Answer (A) Bharatanatyam
V.P. and Shanta Dhananjayan, who trained under Rukmini Devi Arundale at Kalakshetra, are Bharatanatyam dancers and teachers.

Q.17 · Answer (D) Kailasha temple, Kanchipuram
The Kailasanatha temple at Kanchipuram, built by the Pallavas, is in the southern Dravida style. The Konark Sun temple, the Lingaraja temple and Khajuraho's Kandariya Mahadeva temple are Nagara temples.

Q.18 · Answer (A) Hajipur
East Central Railway has its headquarters at Hajipur in Bihar.

Q.19 · Answer (D) Kathak
Pandit Lachhu Maharaj, of the Lucknow gharana, was a Kathak dancer who also choreographed Hindi films.

Q.20 · Answer (B) Shanta Thoutam
Shanta Thoutam, Telangana's Chief Innovation Officer, received the World Innovation Award at the first BRICS Innovation Forum in Moscow, for her contribution to Sustainable Development Goal 4 (quality education).

Q.21 · Answer (B) Gandhi Jayanti
Gandhi Jayanti, on 2 October, is a national holiday marking Mahatma Gandhi's birth. Good Friday, Guru Purab and Muharram are Christian, Sikh and Islamic observances.

Q.22 · Answer (D) 2.9%
SSC's key gives 2.9%. The 2001 Census recorded more than 5 million immigrants, 96 per cent of them from neighbouring countries, chiefly Bangladesh (3.0 million), Pakistan (0.9 million) and Nepal (0.5 million).

Q.23 · Answer (D) Uttarakhand
SSC's key gives Uttarakhand: Abhinav Bindra, India's first individual Olympic gold medallist (10 m air rifle, Beijing 2008), was born in Dehradun. His family is Punjabi, so Punjab is also arguable.

Q.24 · Answer (B) Rajasthan
Both are found in Rajasthan: BWhw (hot desert) in its far west, and BShw (semi-arid steppe) in parts of western Rajasthan, as well as north-western Gujarat and Punjab.

Q.25 · Answer (D) Prakrit
SSC's key gives Prakrit: 'dhamma' is the Prakrit form of Sanskrit 'dharma', the word Ashoka used in his inscriptions. Pali, the language of the Buddhist canon, is itself a Prakrit and also uses 'dhamma', so 'Pali' is arguable too.

English Language and Comprehension

Q.1 · Answer (D) Rahul speaks fluently English
The adverb follows the object, not the verb: 'Rahul speaks English fluently'.

Q.2 · Answer (D) Insolvent
An 'insolvent' person or firm cannot pay its debts. A recluse lives apart from others, and an insurgent rebels against authority.

Q.3 · Answer (D) dedicating
The correct spelling is 'dedicating'.

Q.4 · Answer (C) interrelated
The correct spelling is 'interrelated', with a double 'r'.

Q.5 · Answer (D) bark
'Bark' fits both blanks: a dog barks, and bark is a tree's outer covering.

Q.6 · Answer (A) I wish I had eraser to erase all the mistakes in my life.
'Eraser' is a countable noun and needs an article: 'I wish I had an eraser'.

Q.7 · Answer (B) enjoyed lunch
The lunch is a particular one, the one he gave, so it needs 'the': 'We enjoyed the lunch he gave us.'

Q.8 · Answer (D) invest
One 'invests' a lump sum in a fixed deposit.

Q.9 · Answer (D) Light
The opposite of 'heavy' rain is 'light' rain.

Q.10 · Answer (D) abstracted
'Mesmerised' means held spellbound, with all attention on the music; 'abstracted' means inattentive, with the mind elsewhere.

Q.11 · Answer (D) the most beautiful
'Only' picks out one painting from all the rest, so the superlative is needed: 'Only the most beautiful painting is to be displayed'.

Q.12 · Answer (D) and the piece of cake
Any piece of cake will do, so the indefinite article is needed: 'and a piece of cake'.

Q.13 · Answer (D) because
Being sick is the reason he missed the meeting, so 'because'.

Q.14 · Answer (B) Being good at growing plants
Someone with a 'green thumb' is good at growing plants.

Q.15 · Answer (B) Tired
'Fatigued' means very tired.

Q.17 · Answer (B) Accomplished
'Proficient' means skilled and competent: 'accomplished'.

Q.18 · Answer (C) performace
The correct spelling is 'performance'.

Q.19 · Answer (C) To settle a dispute peacefully
To 'bury the hatchet' is to end a quarrel and make peace.

Q.20 · Answer (A) Verbatim
'Verbatim' means in exactly the same words as the original.

Q.21 · Answer (C) his
Tucker is male ('He searches'), so 'his nose'.

Q.22 · Answer (A) on
A population that is shrinking is 'on the decline'.

Q.23 · Answer (A) number
'A number of species' means several species, among them right whales and killer whales.

Q.24 · Answer (D) based
An organisation is 'based in' a place: Working Dogs for Conservation is based in Three Forks, Montana.

Q.25 · Answer (D) longer
'No longer' means not any more: there are now too many projects for her to keep track of.

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